Enclosure lid: mesh refinement and response

Preliminary, assumption-based analysis. Actual STEP geometry and numerical solves; material, loading and rigid supports are illustrative. No material-strength or safety pass is claimed.

Centre deflection
0.06623 mm
Under 10 N assumed distributed load

Final tetrahedra
7,411
42,564 displacement DOFs

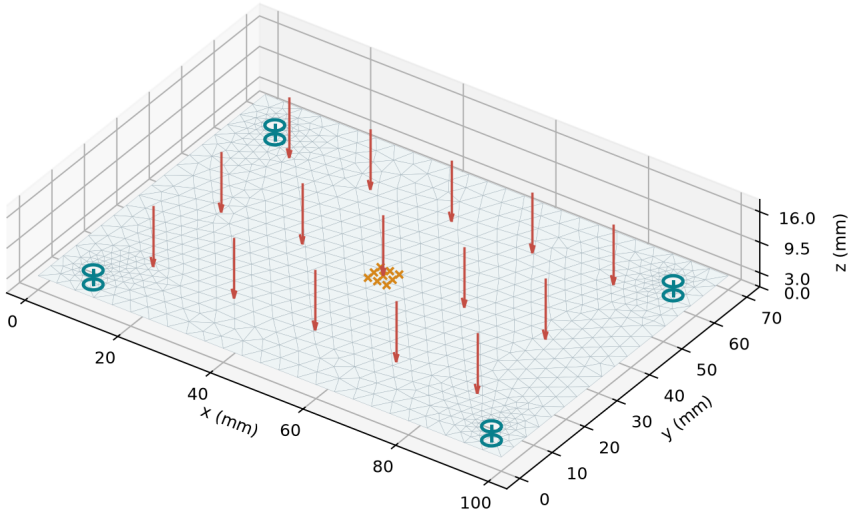
Selected mesh criterion
Met
Two successive comparisons below 2%

1. Geometry and study setup

Named lid extracted from VKA2026.step: 100 × 70 × 3 mm, four 3.5 mm holes. Isotropic material E=2000 MPa, ν=0.35, both assumed. All displacement components fixed on the four screw-hole walls. Uniform downward top-face traction totals 10 N.

The teal marks identify fixed screw-hole supports. Red arrows indicate downward loading. Amber marks the plan locations of nine stress probes inside the upper portion of the lid.

Actual lid geometry and assumed loading/supports



Teal: fixed screw-hole walls. Red: uniform downward (-z) traction; arrow length is schematic.
Total force = 10 N; E = 2000 MPa; nu = 0.35, all assumed.
Amber marks the plan locations of fixed stress probes inside the upper portion of the lid.

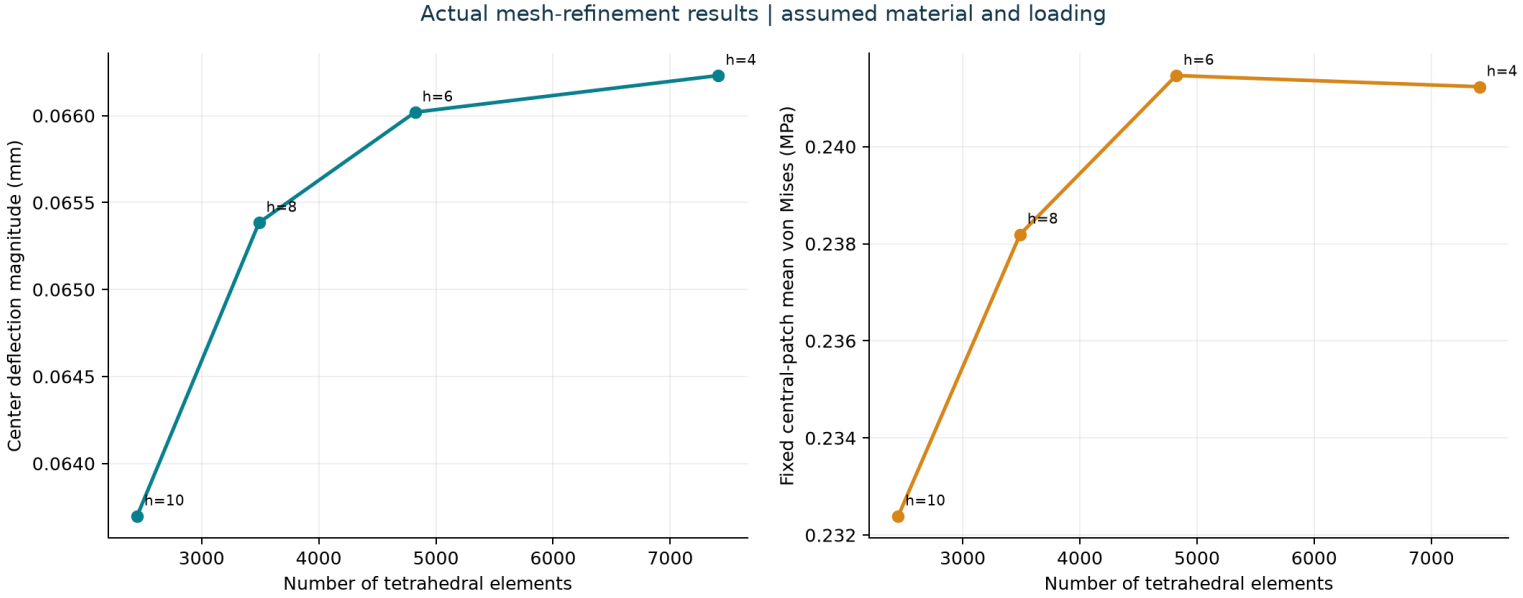
2. Mesh statistics

Target h (mm)	Tetrahedra	Geometric vertices	P2 field nodes	DOFs	Centre abs(uz) (mm)	Probe mean VM (MPa)	Deflection change	Probe stress change
10	2,449	772	4,630	13,890	0.0636966	0.232387	-	-
8	3,490	1,062	6,473	19,419	0.0653845	0.238192	2.581%	2.437%
6	4,822	1,508	9,116	27,348	0.0660195	0.241465	0.962%	1.355%
4	7,411	2,353	14,188	42,564	0.0662305	0.241233	0.319%	0.096%

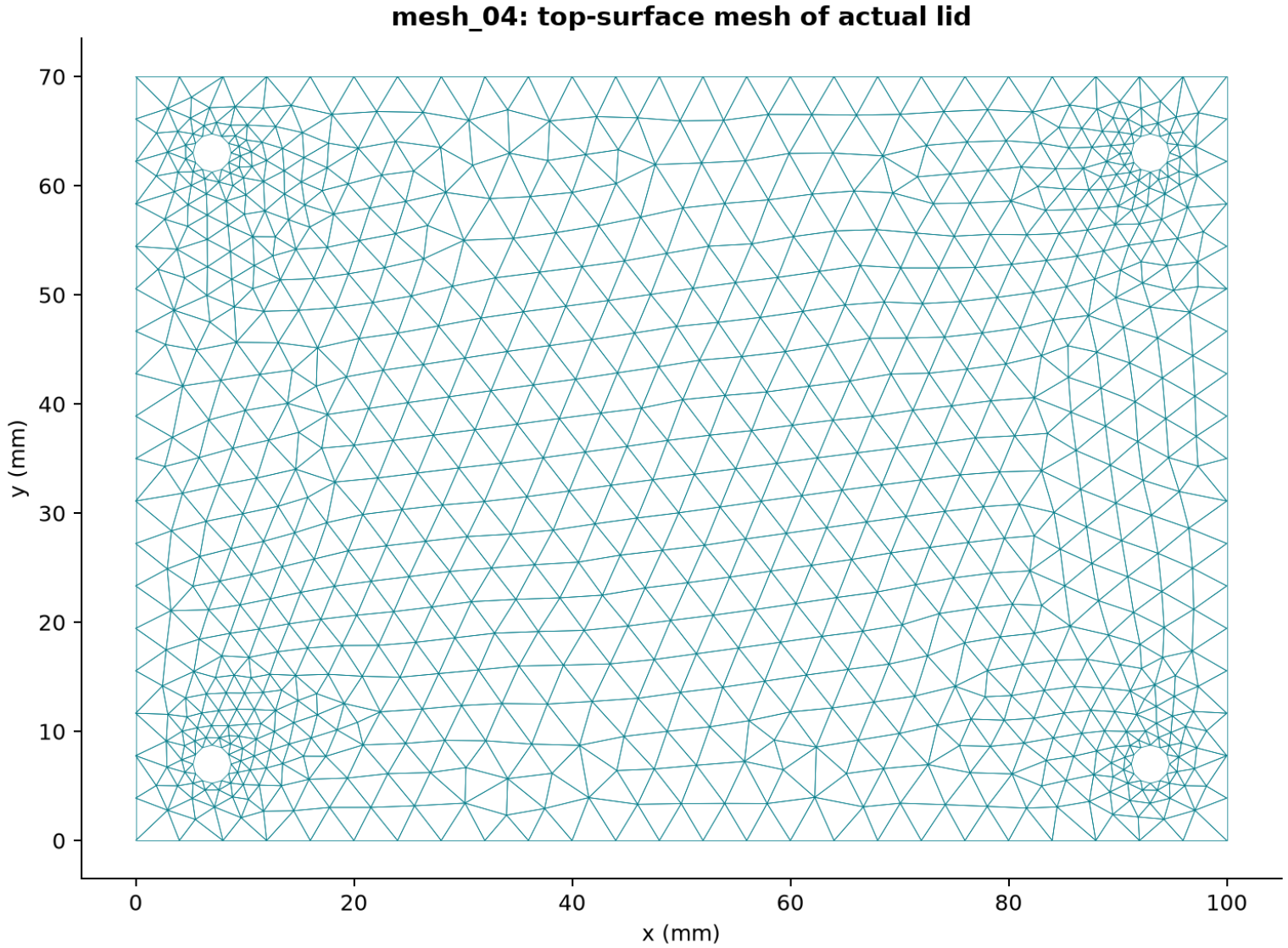
Gmsh stores linear geometric vertices; the solver adds quadratic displacement edge nodes. h is the upper mesh-size control. Curved walls are faceted. Full statistics: CSV.

3. Selected response convergence

Centre displacement and mean von Mises stress at nine fixed central points, z=2.7 mm. These measures changed by 0.319% and 0.096% in the final comparison. This is a successive-mesh criterion, not an absolute error bound.



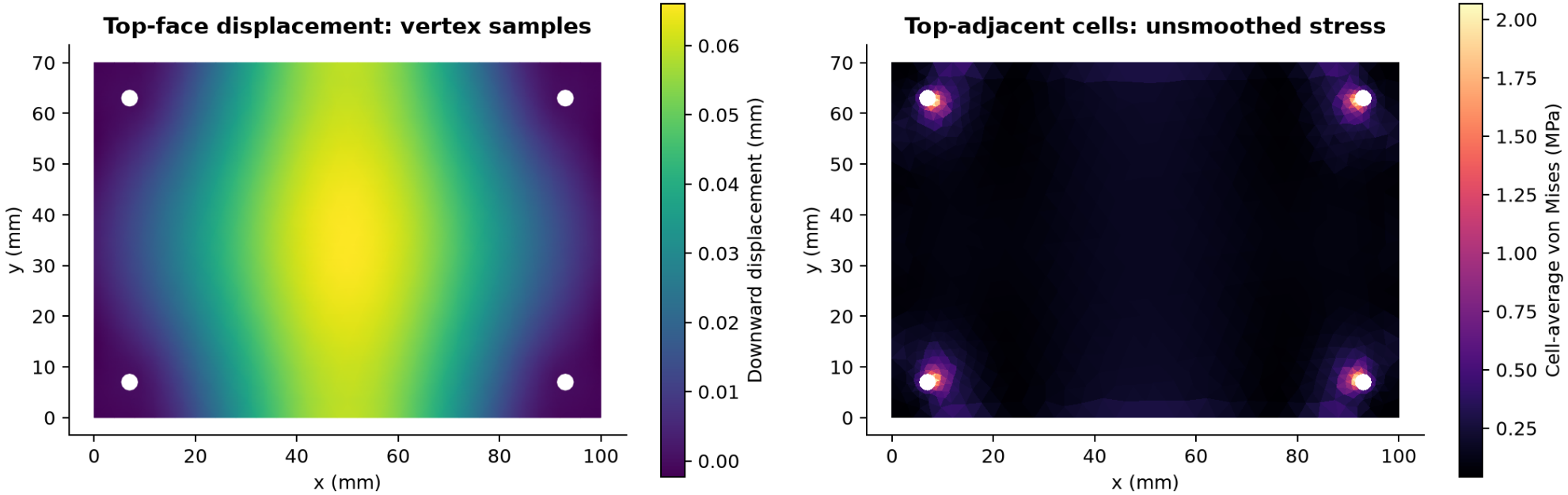
4. Final mesh



7,411 tetrahedra | 2,353 geometric vertices | 14,188 quadratic displacement nodes
Minimum SICN: 0.338 | Straight-sided geometric cells; quadratic displacement field

5. Displacement and stress fields

Preliminary linear-static lid response | no strength pass/fail



Central probe mean: 0.241233 MPa. Largest quadrature-point sample: 2.915480 MPa, with clamp-region peaks that have not been established as converged strength values. No factor of safety is calculated.

6. Numerical checks

The analytical uniaxial-tension patch test passed: relative displacement error 2.420e-15; stress-component error 8.216e-15 MPa. Final lid force-balance error 3.257e-12; free-DOF residual 1.378e-10. These checks verify numerical behavior under the chosen model.

7. Limits and next steps

- Material constants and load are illustrative assumptions, not measured properties or specified service conditions.
- Screw-hole supports are fully fixed; bolts, preload, contact, gasket and enclosure flexibility are not modeled.
- Only the lid is analyzed; electronics, enclosure base, helmet and mounting assembly are excluded.
- Linear, static, small-displacement elasticity only; no impact, drop, creep, fatigue, plasticity, thermal, sealing or intrinsic-safety qualification.
- No material strength has been selected, so no safety factor or strength pass/fail is calculated.
- A successive-mesh comparison is not an absolute error estimate or experimental validation.

The actual enclosure is 100 × 70 × 25 mm; the submission's smaller pod envelope remains a target. Select the real material, manufacturing process and service loads, model actual fastener/contact behavior, and validate with physical tests before making a product decision.

8. Reproducible files

Detailed report and rerun instructions · Editable assumptions · Full results · Analytical check · Final tagged mesh · Field export

Source STEP SHA-256: `8857d052af91eca43b5cf8fd16ea1c35b9f05d296f065b5a0be92e9c094a27be`